

Immunization





Immunogen : substance that stimulate immune response.

- Immunization : inject animal by immunogen to Elicit immune responses.

- Mice and rats are preferred

- Preferred in biomedical research animal model due.

- Anatomical, physiological, and genetic similarity to human.

- Small size.

- Easy to maintenance.

- Short life cycle

- Abundant genetic resources.

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- The preferred species for immunization

mice → Balb/C

Rat → LOU rat

- Why?

- To by the same myloma cells used in fusion process are taken from balb/C and lou rat.



Breeding colony :

Large factories which produce large quantities of Monoclonal Antibodies (grams / weeks)
look for two important things :

1. Cost

2. Convenient

Production of Mab control by different things :

- Immunization: animal that use as source of Antibodies producing cells.
- Feeder cells : release growth factor essential for support Hybridomas growth
- Donor of non immune cells : use as control.
- Production of ascetic fluid : after successful production of target Hybridomas that secrete the target Ab, we take the Hybridomas and inject in mice to allow divide and produce large quantities of Mab.

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- Inside mouse optimal conditions : optimal PH, optimal Temperature, growth factors that will support the growth of Hybridomas and increase the production of Monoclonal antibodies.
 - Sex isn't restriction, using male or female.
 - Mouse old should be 8-12 weeks why?

At this age mouse have strong immune system

- Strong immune system = strong response against Antigens.

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- Billions of Antigens exist, and different from each other some strong and stimulate immune system to Develop strong immune response, other weak lead stimulate weak immune response, other don't stimulate immune response.
 - Certain factors play important role in determine if the degree of immunogenictiy :
 - Foreignness homo polymer like Starch, or poly- Alanin, consist of one type of epitope, Homo polymer are not immunogenic, the immunogen should be heterogeneous.

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- Molecular weight → as molecular weight increase the number of epitopes increase-strong immunogen.
 - Complexity→ more complex structure = more epitopes, 3ry structure more immunogen than 2ry structure.
 - Rout of injection how the antigen enter the body reflect how the body Develop immune response.

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- IP, Intra nasal, IM, orally, SC
 - Antigen should be soluble to be immunogen
 - Protein is the strongest immunogen, then carbohydrates, the lipids.
 - Nucleic acids are poor or non immunogens.

Conjugation :

- Haptens: substance that weak or does not stimulate immune response, after link (Covalent linking not mixing) with carrier protein (conjugate) can stimulate strong immune response.
- Immune system will produce immune response against both hapten and carrier, Antibodies against hapten + Antibodies against carrier.
- Common protein carriers : BSA, KLH, Ig.
- Common coupling Agents : glutaraldehyde and carbodiimide.

Adjuvant :

Substance that increase immune response without change antigen determinants (epitopes).

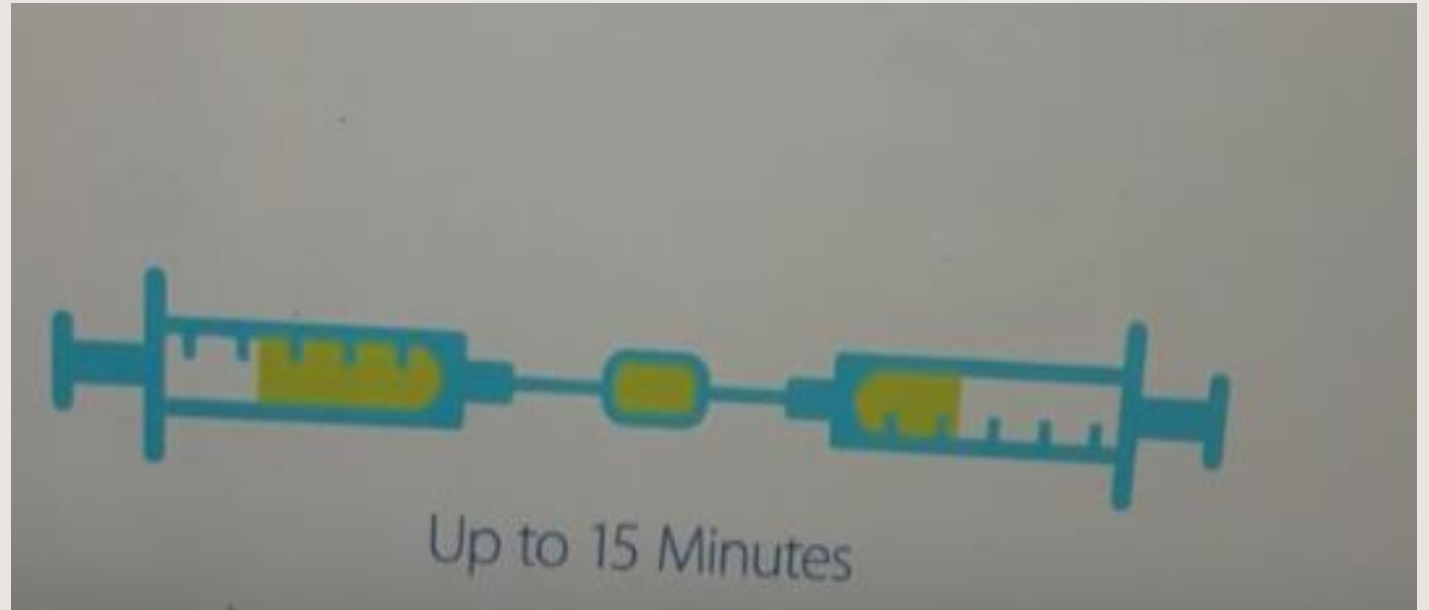
Complete freund Adjuvant (CFA) : oil + water + killed mycobacterium tuberculosis.

Incomplete freund Adjuvant (IFA): oil + water.

Adjuvant principle

Antigen mix with Adjuvant (1:1)→ inject the mouse→antigen adsorb on Adjuvant→ gradually(slow) release of Antigen→ to allow immune system recognize Antigen more than one time→ strong Immune response.

- Adjuvant: allow secondary immune response which strongest than primary immune response.
- Adjuvant: help to Elicit primary and secondary immune response.
- Test the stability of Adjuvant → small drop of Adjuvant in 100 ml beaker of water, if the Adjuvant drop remain discrete droplet and no spread across the surface the Adjuvant is stable.



Immunization procedure :

- After Antigen enter injected the immune response divide into 4 phases

1.Lag phase (Latent phase)→no Antibodies exist in serum.

2.Log phase (exponential phase) → increase production of Antibodies

3.Steady phase→number of Antibodies produce equal number of Antibodies destroy.

4.Decline phase→ immune system shut down.



Lag phase approximately 2 week.

- After second exposure for the same Antigen • secondary immune response.
- Secondary immune response stronger than primary immune response
- Short lag phase, quicker, last longer, and higher affinity.
- Optimal dose : 10 – 100 microgram.
- Purity : Antigen not necessarily be pure, using pure or no pure Antigen.

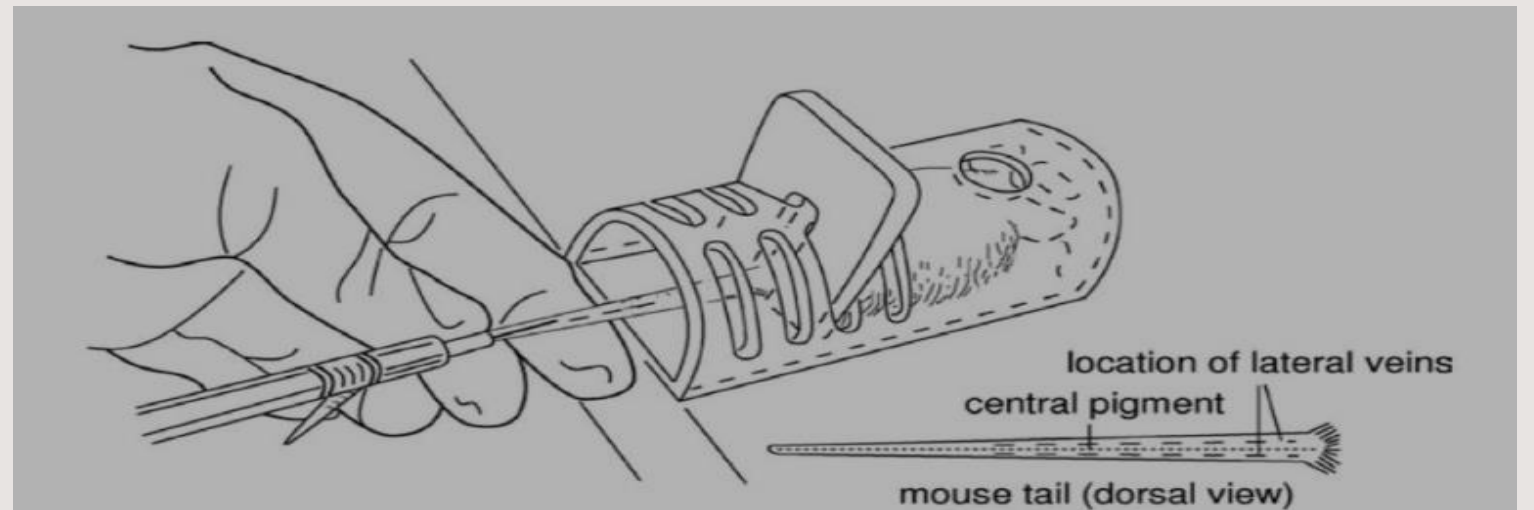
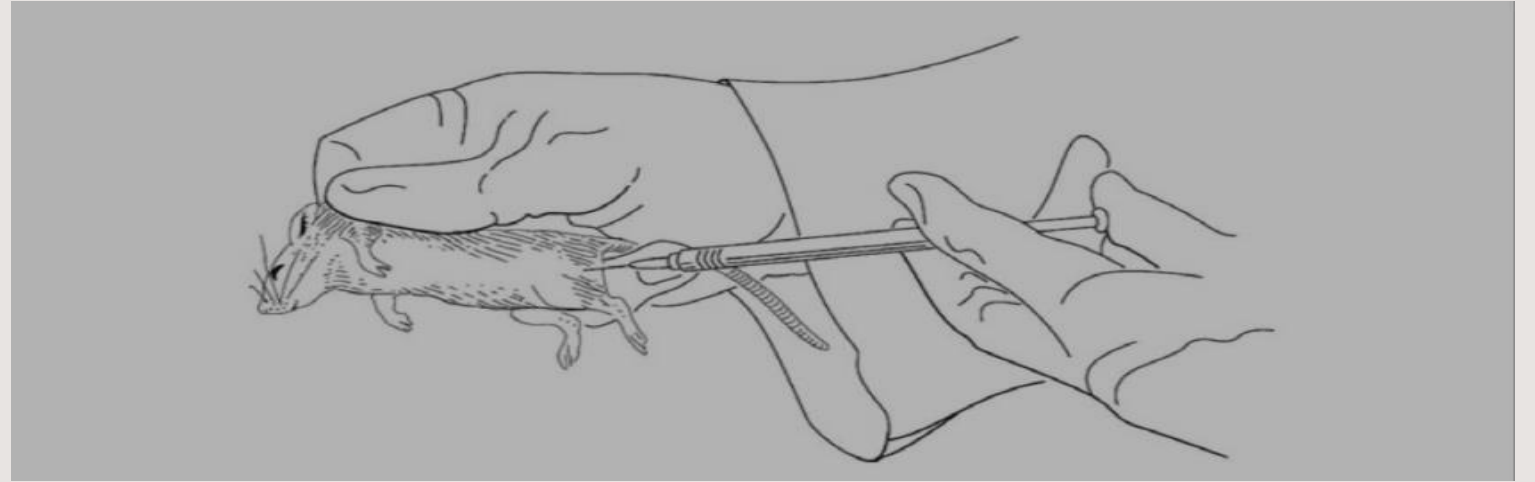
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- Target Antigen: specific B cells secrete Antibodies → isolation of specific B cells and fused with myeloma cell to produce Hybridomas.
 - Why the B cells (plasma cells) not use directly to Monoclonal production?

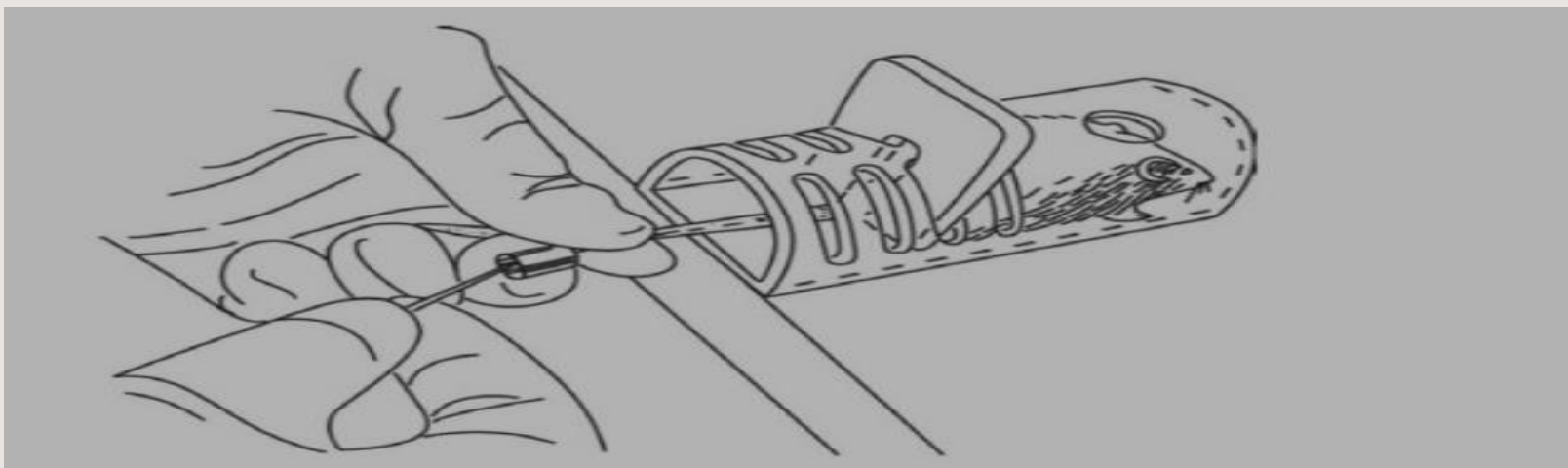
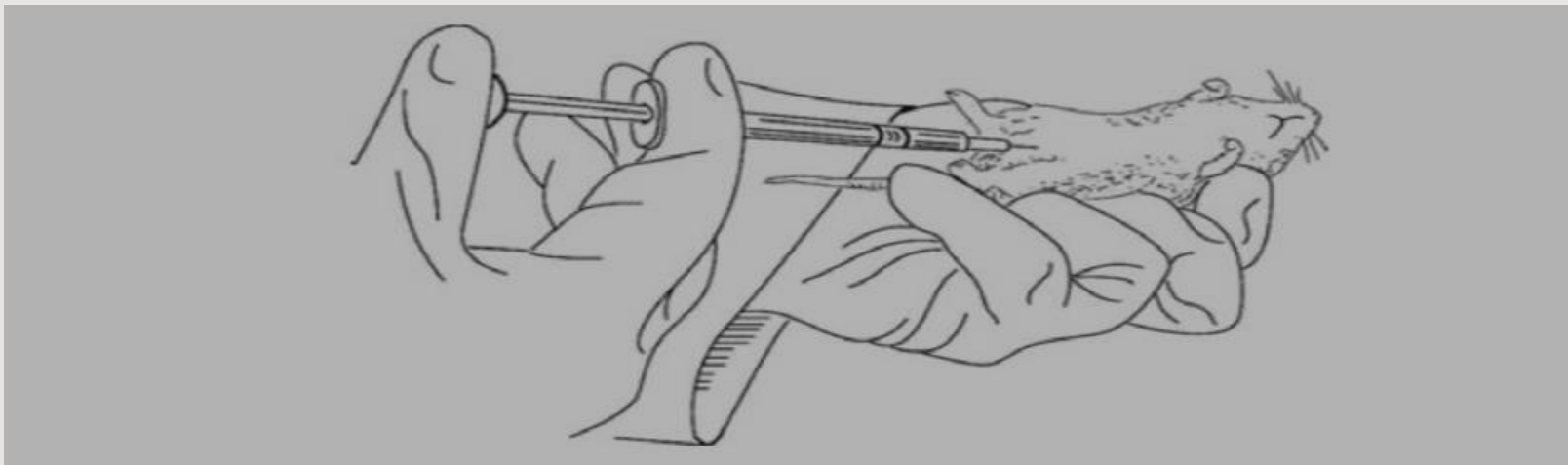
Plasma cells have limit life span few days then die, but Hybridomas immortal can divide indefinitely and secrete antibodies.

- First dose of antigen inject, and after 3-4 day secondary dose inject (booster injection).

Routes of administration :

- Aim of using Adjuvant: slow release of Ag, we choose site where not highly vascular area.
- Intra venues injection not preferred→ Ag will eliminate and metabolize quickly, also high risk of animal shock.
- The sites preferred : IP, SC
- Positive control (using Rabbit) : IM multiple injections and IV booster doses.





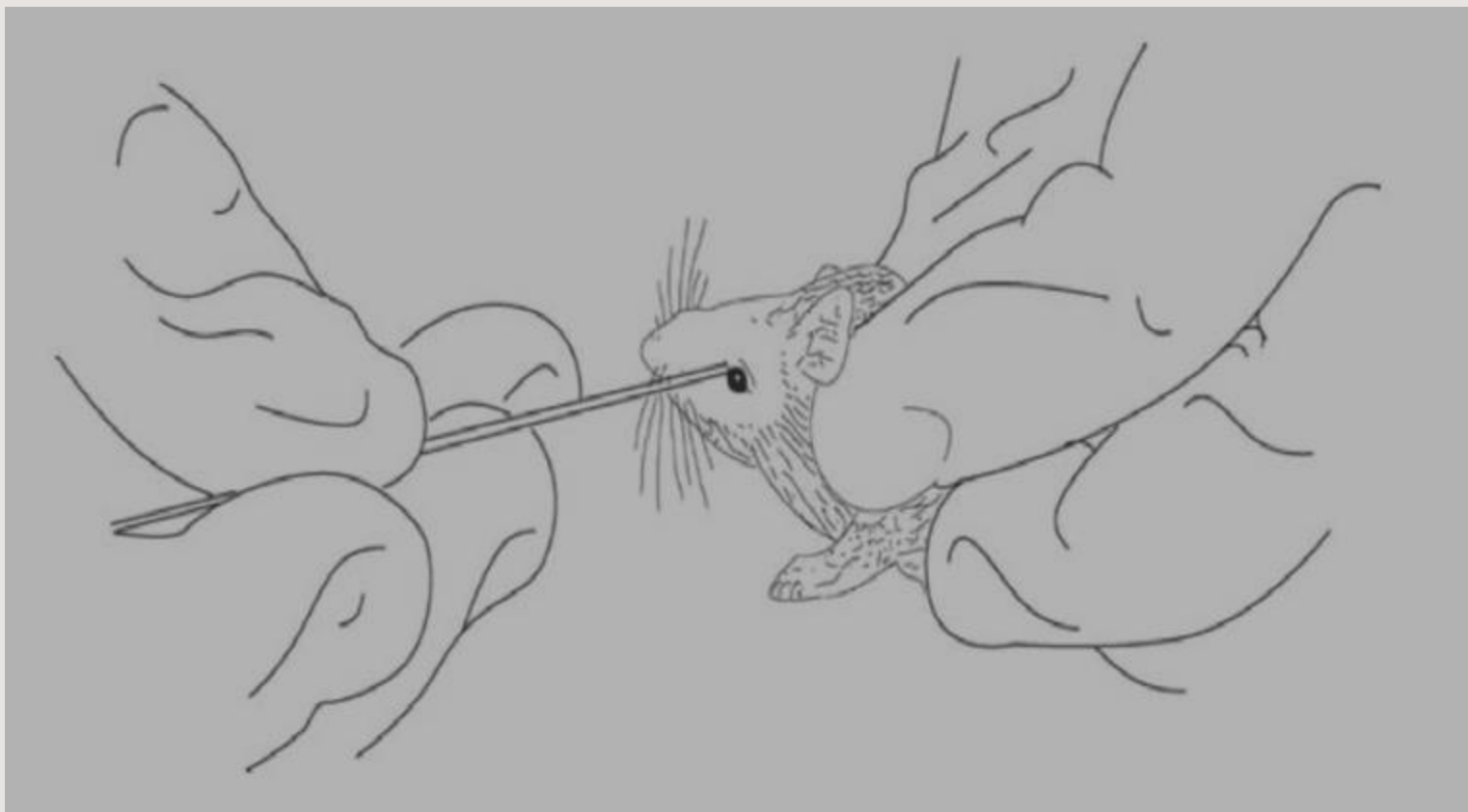
Bleeding methods

- Mice : tail or retro-bulber (from eyes) or saphenous vein.
- Rabbit : marginal ear zone

In vivo immunization

Inject animal with target antigen.

- Tolerance: animal could tolerant with the injected Ag and does not Develop immune response.
- Large quantities of Antigen needed.
- Overwhelming response to other Antigens
- Ag could be toxic to animal.



In vitro immunization

- Cell Culture contain B cells + target Antigen.
- No need animal model.
- No sensitization to Ags occurs if one uses PBMCs directly isolated from human blood. Treatment of PBMCs with L-leucyl-L-leucine methyl ester (LLME) removes
 - these immunosuppressive cells and thus enables the in vitro sensitization with any Ag.
- Then, LLME treated cells are immunized with Ags in the presence of cytokines (IL-2, IL- 4), CpG or addition of Anti IL-40, IL-7 and LPS in tissue culture.

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- LLME is a lysosomal condensation product that has been reported to be cytotoxic towards NK cells and CD8+ T lymphocytes without affecting helper T cells and B cells.
 - It has also been shown to induce death of monocytes, polymorphonuclear leukocytes, and myeloid tumor cells.
 - The stimulated B lymphocytes were transformed into permanently Ab- producing hybridomas by cell fusion technology followed by selection and cloning.

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- Compared to the control, PBMCs immunized in vitro demonstrate an augmentation of total Ab productions, and induced Ag-specific Ab producing cells .

Procedure :

- Spleen cell Suspension+ complete culture media + Ag+ Muramly dipeptide (MDP) → (recommend Adjuvant in in vitro immunization)